Field of computer Science & Engineering

Data Mining

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Behind the Stars: Mining Customer Reviews for Restaurant Hygiene Using Fuzzy Logic		
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Notice: This part for instructor		**Degree**
Date of Submission	**29/12/2025**	

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Abstract:

Online restaurant reviews contain valuable information about hygiene and cleanliness that is often overlooked when customers rely only on star ratings. While star ratings usually reflect food quality and service, hygiene-related concerns are frequently expressed in review text rather than numerical scores.

In this project, we propose a data mining approach to evaluate restaurant hygiene by analyzing customer reviews collected from Google Maps. The reviews were pre-processed and analyzed to extract hygiene-related keywords and sentiments. A fuzzy logic inference system was then used to handle the uncertainty of natural language and combine these signals into a single, interpretable hygiene score for each restaurant.

The results show that restaurants with high star ratings may still receive low hygiene scores, revealing potential risks that are not visible from ratings alone. This demonstrates that mining textual reviews can provide meaningful hygiene insights that complement traditional rating systems and support more informed decision-making.

Introduction:

Online review platforms play a major role in shaping customer decisions in the restaurant industry. Many customers rely heavily on star ratings to judge restaurant quality, if higher ratings indicate better overall conditions. However, star ratings usually reflect a combination of factors such as food taste, service quality, and price, while important aspects like hygiene and cleanliness are often underrepresented.

Hygiene is a critical factor that directly affects customer health and safety. Complaints related to cleanliness, food safety, and sanitation are frequently expressed in textual reviews rather than numerical ratings. As a result, restaurants with potential hygiene issues may still appear highly rated, making it difficult for customers to accurately assess hygiene conditions based on star ratings alone.

This project aims to address this limitation by mining customer review text to extract hygiene-related information. By focusing on hygiene-specific keywords and sentiments, the system isolates cleanliness signals that are usually hidden within unstructured text. To handle the uncertainty and subjectivity of natural language, a fuzzy logic approach is applied to combine these signals into a single, interpretable hygiene score for each restaurant. This approach provides a clearer and more realistic assessment of restaurant hygiene that complements traditional star rating systems.

Methodology:

This project follows a structured data mining pipeline to evaluate restaurant hygiene using customer review text. The methodology consists of data collection, text preprocessing, hygiene sentiment extraction, fuzzy logic modelling, and hygiene score aggregation.

Data Collection

Customer reviews were collected from Google Maps for a set of restaurants. Each review includes a star rating and a textual comment provided by customers. Since hygiene-related information is often expressed in text rather than numerical ratings, the analysis focuses primarily on review content rather than star scores.

Text Preprocessing

The collected reviews were cleaned and pre-processed to improve text quality and consistency. This process included converting text to lowercase, removing punctuation and irrelevant symbols, and eliminating common stop words. These steps reduce noise and ensure that hygiene-related terms can be accurately identified during analysis.

Hygiene Keyword Identification

A predefined dictionary of hygiene-related keywords was used to detect cleanliness and food safety mentions within reviews. The keywords were divided into positive hygiene indicators (e.g., clean, hygienic, fresh) and negative hygiene indicators (e.g., dirty, unhygienic, contaminated). Reviews containing at least one hygiene-related keyword were classified as hygiene-related reviews and selected for further analysis.

Hygiene Sentiment Scoring

For each hygiene-related review, a hygiene sentiment score was computed based on the balance between positive and negative hygiene keywords. The score is defined as:

$$\boldsymbol{Hygiene\ Sentiment} = \frac{\boldsymbol{N_{pos} - N_{neg}}}{\boldsymbol{N_{pos} + N_{neg}}}$$

where N_{pos} represents the number of positive hygiene keywords and N_{neg} represents the number of negative hygiene keywords in a review. This normalized score ranges from −1 to 1, where higher values indicate more positive hygiene sentiment.

Fuzzy Logic Modelling

Because customer opinions are subjective and expressed in natural language, fuzzy logic was used to model uncertainty in hygiene sentiment. The hygiene sentiment scores were mapped into fuzzy membership functions representing three linguistic categories: **Poor**, **Average**, and **good** hygiene. A fuzzy inference system was then applied to combine these memberships and infer an overall hygiene assessment for each review.

Hygiene Score Aggregation

After defuzzification, each hygiene-related review was assigned a hygiene score between 0 and 1. To obtain a restaurant-level hygiene score, the scores of all hygiene-related reviews associated with a restaurant were averaged. This final hygiene score provides an interpretable measure of restaurant cleanliness that can be compared with traditional star ratings.

Related work:

Previous research has extensively explored the use of online customer reviews to analyze public opinion and service quality in the restaurant industry. Early studies focused on opinion mining and sentiment analysis techniques to classify reviews as positive or negative, demonstrating the value of textual feedback beyond numerical ratings. These approaches showed that review text contains rich information that can influence customer decisions and business performance.

More recent studies have applied sentiment analysis to restaurant reviews using machine learning and lexicon-based methods. While these techniques are effective in capturing overall sentiment, they often treat reviews as a single polarity score and do not distinguish between specific quality dimensions such as hygiene, cleanliness, or food safety. As a result, important hygiene-related concerns may remain hidden within general sentiment classifications.

Fuzzy logic has been widely used in decision support systems and service quality evaluation due to its ability to handle uncertainty and linguistic ambiguity. Unlike traditional classification methods, fuzzy logic allows gradual transitions between categories and better represents subjective human judgments. Several studies have successfully applied fuzzy logic to evaluate service quality, customer satisfaction, and risk assessment in uncertain environments.

Building on these works, this project combines text mining techniques with fuzzy logic to specifically evaluate restaurant hygiene. By focusing on hygiene-related keywords and modelling uncertainty through fuzzy inference, the proposed approach provides a more interpretable and domain-specific hygiene assessment compared to traditional sentiment analysis and star rating systems.

Comparison between methods:

Several approaches can be used to assess restaurant hygiene from online reviews:

1. **Star Rating Analysis**

   Star ratings provide a simple numerical summary but do not isolate hygiene factors and often mix multiple service dimensions.

2. **Traditional Sentiment Analysis**

   Polarity-based sentiment analysis identifies positive and negative opinions but lacks specificity regarding hygiene and struggles with linguistic ambiguity.

3. **Proposed Fuzzy Logic-Based Method**

   The proposed method extracts hygiene-related keywords, distinguishes between positive and negative hygiene indicators, and applies fuzzy membership functions to model uncertainty. This approach produces an interpretable hygiene score that better reflects real customer experiences.

Compared to traditional methods, the fuzzy logic approach provides **higher interpretability, flexibility, and domain specificity**.

Recommendation:

Based on the findings of this project, several practical recommendations can be made for different stakeholders.

First, online restaurant review platforms should consider integrating text-based hygiene analysis alongside traditional star ratings. Presenting a hygiene score derived from customer reviews would help users make more informed decisions by highlighting cleanliness and food safety concerns that are not reflected in numerical ratings.

Second, restaurant managers can use hygiene scores extracted from customer feedback as an early warning indicator. By monitoring hygiene-related complaints in reviews, managers can identify cleanliness issues, improve sanitation practices, and respond proactively to customer concerns before they escalate.

Third, the proposed approach can be extended and improved in future work by expanding the hygiene keyword dictionary and incorporating more advanced natural language processing techniques. Combining fuzzy logic with machine learning models may further enhance the accuracy and robustness of hygiene assessment.

Finally, regulatory bodies and food safety inspectors may benefit from automated hygiene monitoring tools based on customer reviews. Such systems could support inspection prioritization by identifying restaurants with consistently low hygiene scores, helping authorities allocate resources more effectively.

Results:

The proposed hygiene evaluation approach was applied to a dataset of customer reviews collected from Google Maps. In total, 6,552 reviews were analysed, out of which 860 reviews contained hygiene-related keywords and were classified as hygiene-related reviews. These reviews were used to compute hygiene sentiment scores and final hygiene scores for each restaurant.

The analysis revealed noticeable variation in hygiene scores across restaurants. While some restaurants achieved consistently high hygiene scores, several restaurants exhibited low hygiene scores despite having relatively high star ratings. This indicates that hygiene-related concerns are not always reflected in numerical ratings and are often embedded within review text.

To examine the relationship between star ratings and hygiene scores, a correlation analysis was conducted. The results showed a weak positive correlation (approximately 0.36) between star ratings and hygiene scores. This suggests that star ratings alone are not a reliable indicator of restaurant hygiene and cleanliness.

In several cases, restaurants with high average star ratings received low hygiene scores due to repeated mentions of negative hygiene-related keywords such as poor cleanliness or food safety issues. These cases highlight the importance of analysing textual reviews to uncover hygiene risks that may be overlooked when relying solely on star ratings.

Overall, the results demonstrate that the proposed fuzzy logic-based approach successfully captures hygiene-related insights from customer reviews and provides a more detailed and interpretable assessment of restaurant hygiene compared to traditional rating systems

Power PI Graphs:

Figure 1: Case Study

This case study highlights the limitation of relying solely on star ratings to evaluate restaurant hygiene. Although *New York Fried Chicken* has a high average star rating of **4.40**, its hygiene score is relatively low (**0.59**). The most frequent hygiene-related keyword in customer reviews is *"dirty"*, indicating repeated cleanliness concerns. This example demonstrates how negative hygiene signals embedded in review text can reveal risks that are not reflected in numerical ratings.

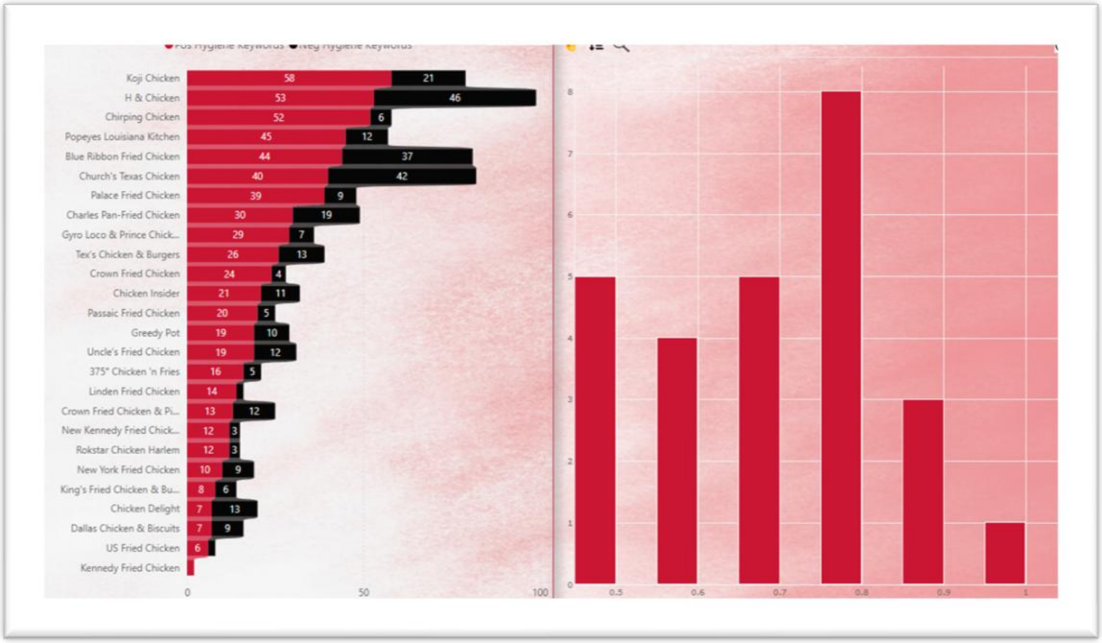

Figure 2: Hygiene Keyword Frequency by Restaurant

This chart shows the frequency of hygiene-related keywords across different restaurants. Some restaurants exhibit a high number of negative hygiene mentions, while others show fewer occurrences. The variation indicates that hygiene concerns are not uniformly distributed and that certain restaurants consistently attract more cleanliness-related complaints. This supports the need for text-based hygiene analysis rather than relying on aggregated star ratings alone.

Figure 3: Distribution of Hygiene Scores

The histogram illustrates the distribution of hygiene scores across the analysed restaurants. Most hygiene scores fall within the mid-range, while a smaller number of restaurants achieve very high or very low scores. This distribution reflects varying hygiene conditions among restaurants and confirms that hygiene quality differs significantly even within the same category of establishments.

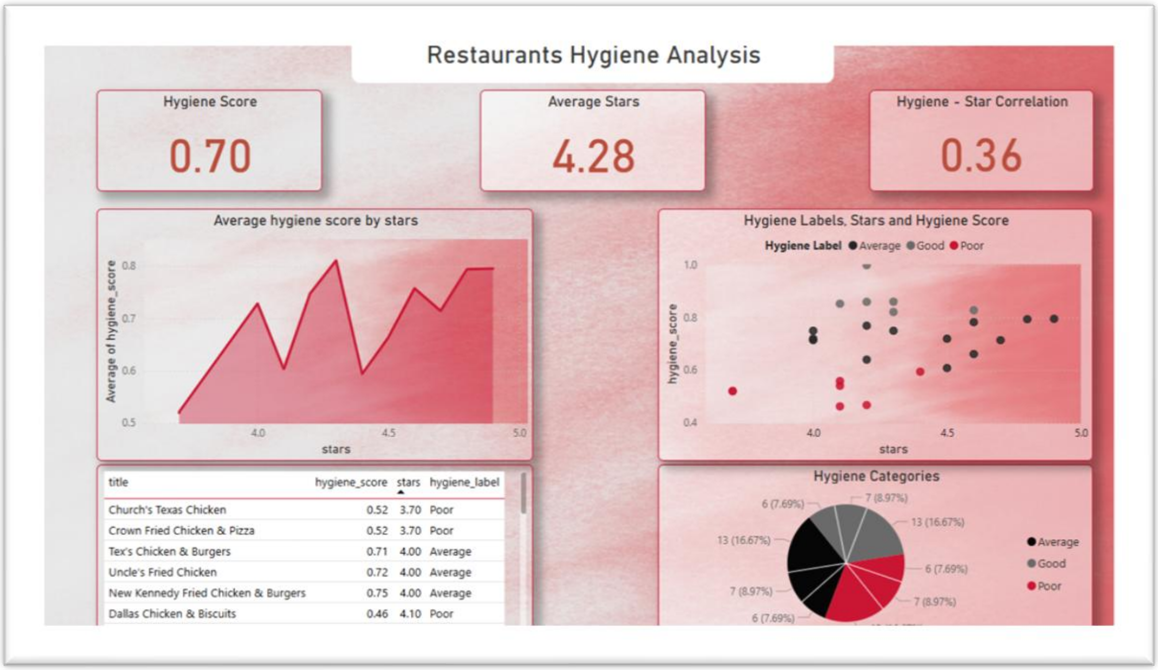

Figure 4: Distribution of Hygiene Scores

The histogram illustrates the distribution of hygiene scores across the analyzed restaurants. Most hygiene scores fall within the mid-range, while a smaller number of restaurants achieve very high or very low scores. This distribution reflects varying hygiene conditions among restaurants and confirms that hygiene quality differs significantly even within the same category of establishments.

Figure 5: Restaurants Hygiene Analysis Dashboard

(a) Average Hygiene Score

The average hygiene score across all restaurants is approximately 0.70, indicating generally acceptable hygiene conditions, while still leaving room for improvement in several establishments.

(b) Average Star Rating

The average star rating is 4.28, showing that most restaurants are rated highly by customers. However, this high average contrasts with the variation observed in hygiene scores.

(c) Hygiene–Star Correlation

The correlation between star ratings and hygiene scores is approximately 0.36, indicating a weak positive relationship. This suggests that star ratings alone are not reliable indicators of restaurant hygiene.

(d) Hygiene Score vs. Star Rating Scatter Plot

The scatter plot shows that restaurants with similar star ratings can have very different hygiene scores. This further confirms that hygiene information is not consistently captured by star ratings.

(e) Hygiene Category Distribution

The pie chart categorizes restaurants into *Good*, *Average*, and *Poor* hygiene groups. The presence of restaurants in all categories highlights the diversity of hygiene conditions and supports the usefulness of categorical hygiene labelling for clearer interpretation.

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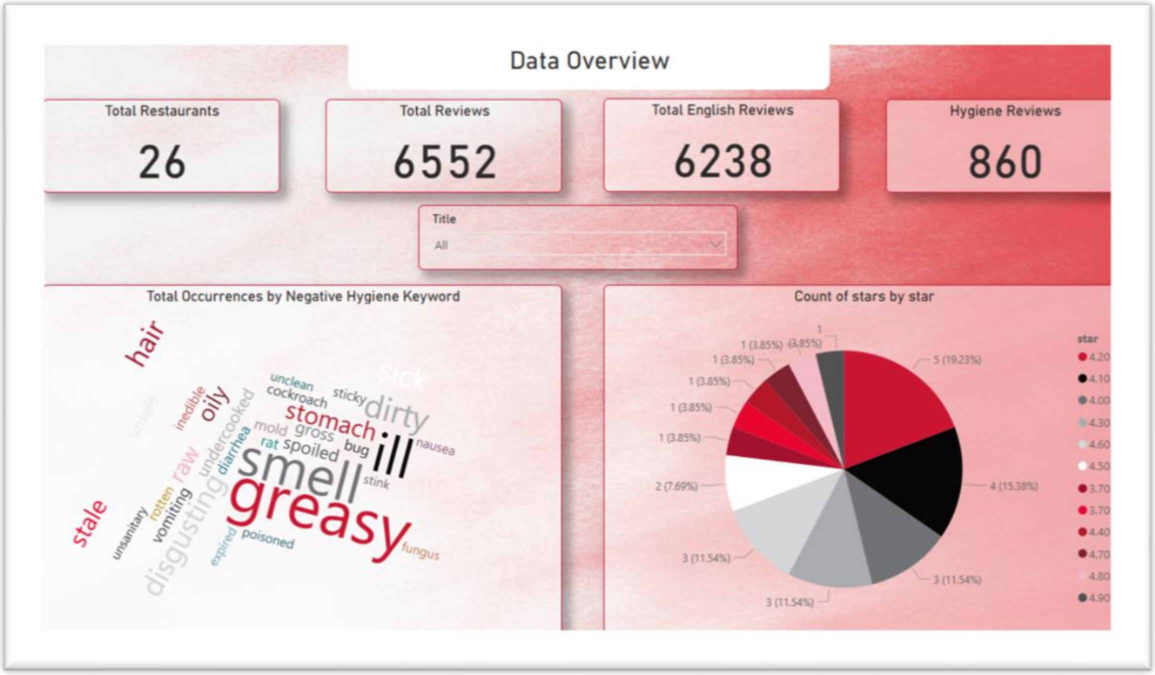

Figure 6: Data Overview Dashboard

(a) Dataset Summary

Comment:

The dataset consists of 26 restaurants and 6,552 total reviews, of which 6,238 are English reviews. A total of 860 reviews were identified as hygiene-related and used in the hygiene analysis.

(b) Negative Hygiene Keyword Cloud

Comment:

The word cloud visualizes the most frequent negative hygiene-related keywords, such as *"smell," "greasy," "dirty,"* and *"sick."* These terms reflect common customer concerns related to cleanliness and food safety.

(c) Star Rating Distribution

Comment:

The star rating distribution shows that most reviews fall within the higher rating range. Despite this, hygiene-related complaints are still present, reinforcing the finding that high star ratings do not necessarily imply good hygiene conditions.

Conclusion:

This project presented a data mining approach for evaluating restaurant hygiene by analysing customer review text using fuzzy logic. By moving beyond traditional star ratings and focusing on hygiene-related keywords embedded in reviews, the proposed system was able to uncover cleanliness and food safety concerns that are often overlooked in numerical ratings.

The results demonstrated that restaurants with high star ratings may still exhibit low hygiene scores, and the observed weak correlation between star ratings and hygiene scores confirms that ratings alone are not reliable indicators of hygiene conditions. The fuzzy logic framework effectively handled the uncertainty and subjectivity of natural language, producing interpretable hygiene scores that better reflect customer experiences.

Overall, this work highlights the value of mining unstructured review text to support more informed decision-making for customers and restaurant managers. Future improvements may include expanding the hygiene keyword dictionary, incorporating advanced natural language processing techniques, and applying the approach to larger datasets and real-time review streams.

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